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Diagnosis

VISION IS THE BOTTLENECK

- MLLM progress often comes from stronger LLMs while vision encoder choices remain underexplored.
- Several popular benchmarks can be answered largely without images and contain strong language bias.
- Testing 23 vision backbones reveals distinct General, Knowledge, OCR/Chart and Vision-Centric clusters.

2

CV-Bench

FUNDAMENTAL 2D + 3D

- 2,638 manually checked VQA examples repurpose COCO, ADE20K and Omni3D annotations.
- 2D tasks test spatial relationships and counting; 3D tasks test depth order and relative distance.
- It is 3.5x larger than RealWorldQA and 8.8x larger than MMVP.

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Vision findings

WHAT ENCODERS TEACH US

- Two-stage training and more adapter data help; unfreezing the vision encoder is broadly beneficial.
- Language-supervised encoders lead overall, especially OCR, but high-resolution DINOv2 excels on vision-centric tasks.
- Combining SigLIP, CLIP, DINOv2 and ConvNeXt yields complementary representations.

CAMBRIAN-1

A Fully Open, Vision-Centric Exploration of MLLMs

Vision Encoders, SVA, CV-Bench & Curated Data | 2024

Use multimodal instruction tuning as a new evaluator for visual representations, then co-design benchmark, vision connector and data to strengthen sensory grounding in

MLLM

open MLLMs. visual grounding

open cookbook

4

SVA

SPATIAL VISION AGGREGATOR

- Learnable latent queries cross-attend to multiple encoders at their native resolutions.
- Spatial inductive bias restricts each query to a defined image region and preserves layout.
- SVA is inserted repeatedly across LLM layers, letting language revisit visual evidence instead of seeing it once.

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Data & results

QUALITY OVER RAW SCALE

- Cambrian-10M is balanced and filtered into higher-quality Cambrian-7M; average score rises 54.8 -> 55.9.
- Cambrian-1 combines four encoders, 2.5M adapter samples and Cambrian-7M instruction tuning.
- The 8B/13B/34B family beats leading open MLLMs and uses only 576 visual tokens - about one-fifth of peers.

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Meaning & limits

OPEN RESEARCH STACK

- The release includes models, code, CV-Bench, datasets and large-scale training/evaluation recipes.
- It warns against 'answer machine' optimization: benchmarks do not capture conversation, reliability or user experience.
- Ultra-high resolution, extreme aspect ratios, RL alignment and better data curation remain future work.